DATA SCIENCE & ANALYTICS INTERNSHIP

Task 2 : Customer Retention and Churn Analysis

Submitted by : Kripa K

Internship : Future Interns

Domain : Data Science & Analytics

1. Introduction

Customer retention is an important aspect of business success, as retaining existing customers is often more cost-effective than acquiring new ones. Customer churn occurs when customers stop using a company's products or services. Understanding why customers leave helps businesses improve customer satisfaction, strengthen customer relationships, and increase long-term profitability.This project focuses on analyzing customer churn using the Telco Customer Churn dataset. Python was used for data cleaning, exploratory data analysis, and visualization, while Microsoft Power BI was used to create an interactive dashboard for presenting the findings.

Dataset Description

Dataset name: Telco Customer Churn Dataset
Source: Kaggle
Domain: Telecommunications / Customer Analytics
Number of records: 7,043 customer records
Purpose: To analyze customer churn patterns, identify factors associated with customer cancellations, and understand customer retention behavior to support data-driven business decisions.

Problem Statement

Telecommunications companies face challenges in retaining customers because of differences in customer needs, service experiences, contract commitments, and pricing. Without proper analysis, it can be difficult to identify customer groups with a higher likelihood of leaving.This project addresses the problem by analyzing historical customer data to identify churn patterns and potential risk factors. The findings can help businesses design targeted retention strategies and make data-driven decisions.

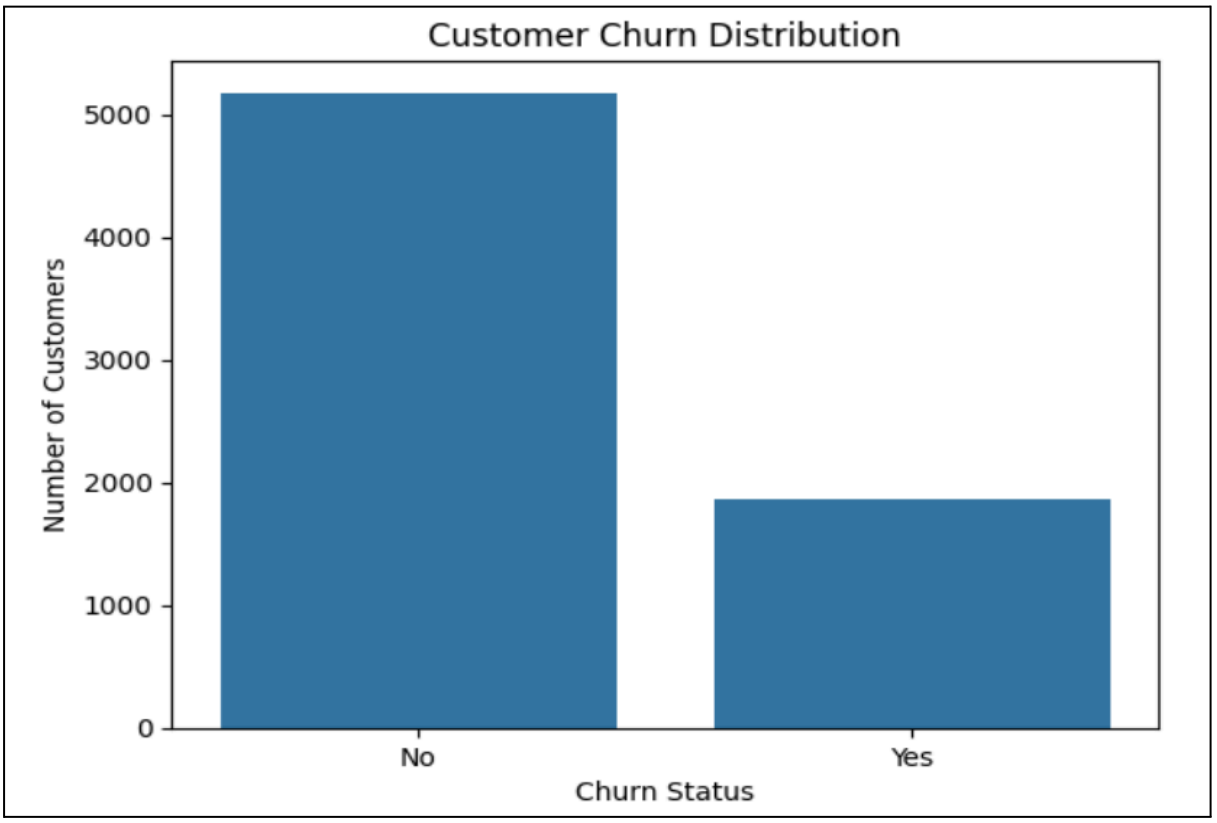
Project Overview

The project follows a structured data analysis workflow. The dataset was imported into Jupyter Notebook, cleaned, and prepared for analysis. Exploratory data analysis was performed using Python libraries to examine churn patterns across important customer attributes.The results were then presented through charts and a Microsoft Power BI dashboard. Finally, key findings and business recommendations were developed to support customer retention and improve decision-making.

2. Data Analysis and Visualization

2.1 Customer Churn Distribution

Graph: Customer Churn Count



Interpretation:

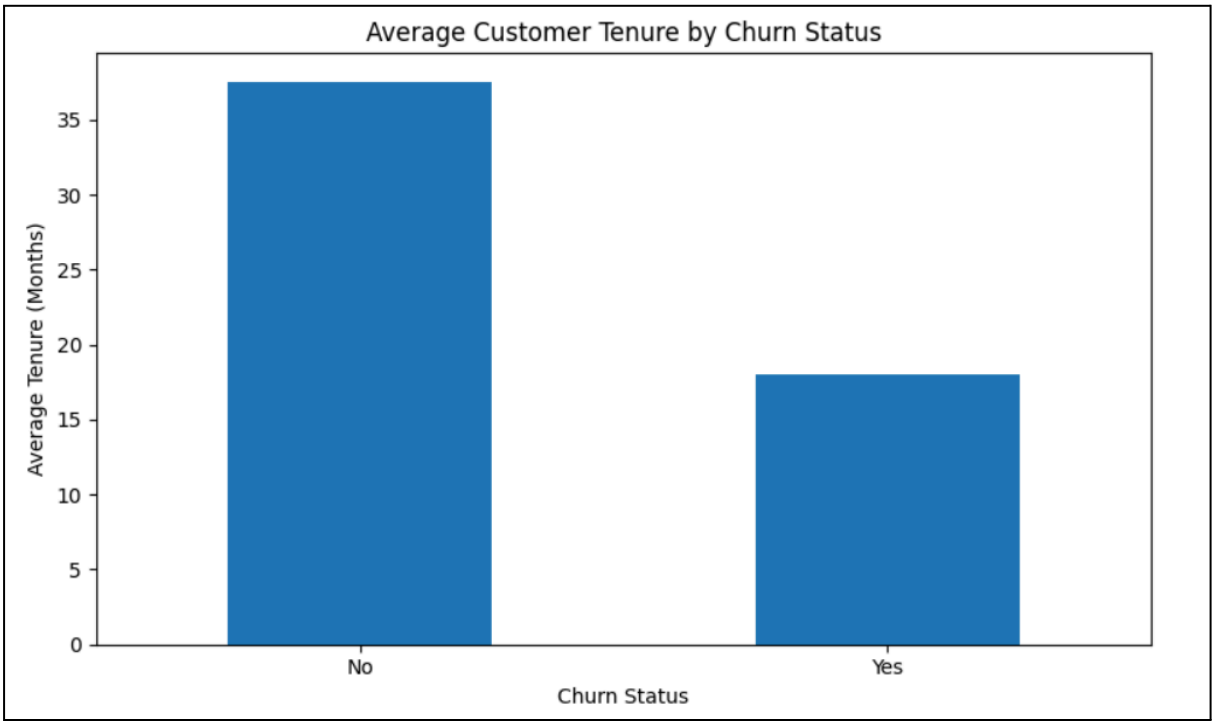
The dataset contains 7,043 customers, of whom 1,869 customers have churned and 5,174 customers have remained with the company. The overall churn rate is approximately 26.54%, while the retention rate is 73.46%.

Business Insight:

Although most customers remain with the company, more than one-quarter of customers have left. This indicates a need for targeted customer retention strategies.

2.2 Customer Tenure by Churn Status

Graph: Customer Tenure by Churn Status



Interpretation

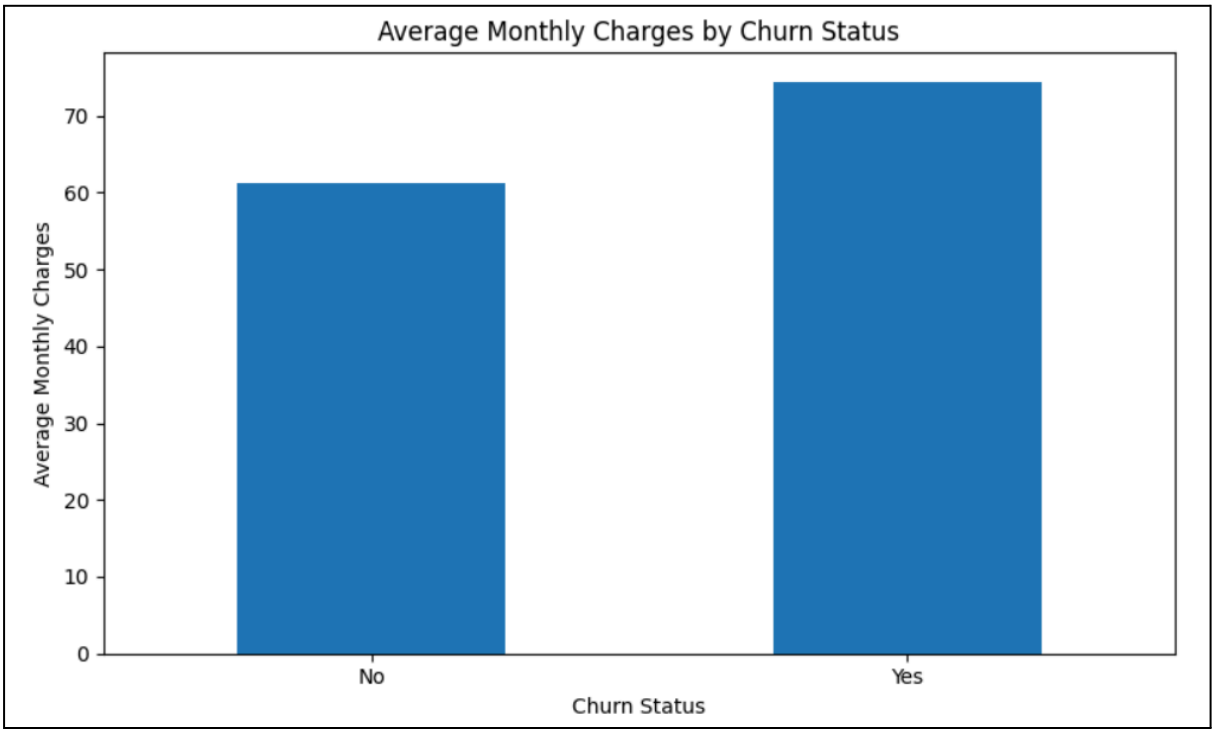
Customers who churned had an average tenure of approximately 18 months, compared with approximately 38 months for customers who stayed.

Business Insight

Customer churn is associated with shorter customer tenure. Strengthening onboarding, early engagement, and customer support may help reduce early customer losses.

2.3 Monthly Charges by Churn Status

Graph: Monthly Charges by Churn Status



Interpretation

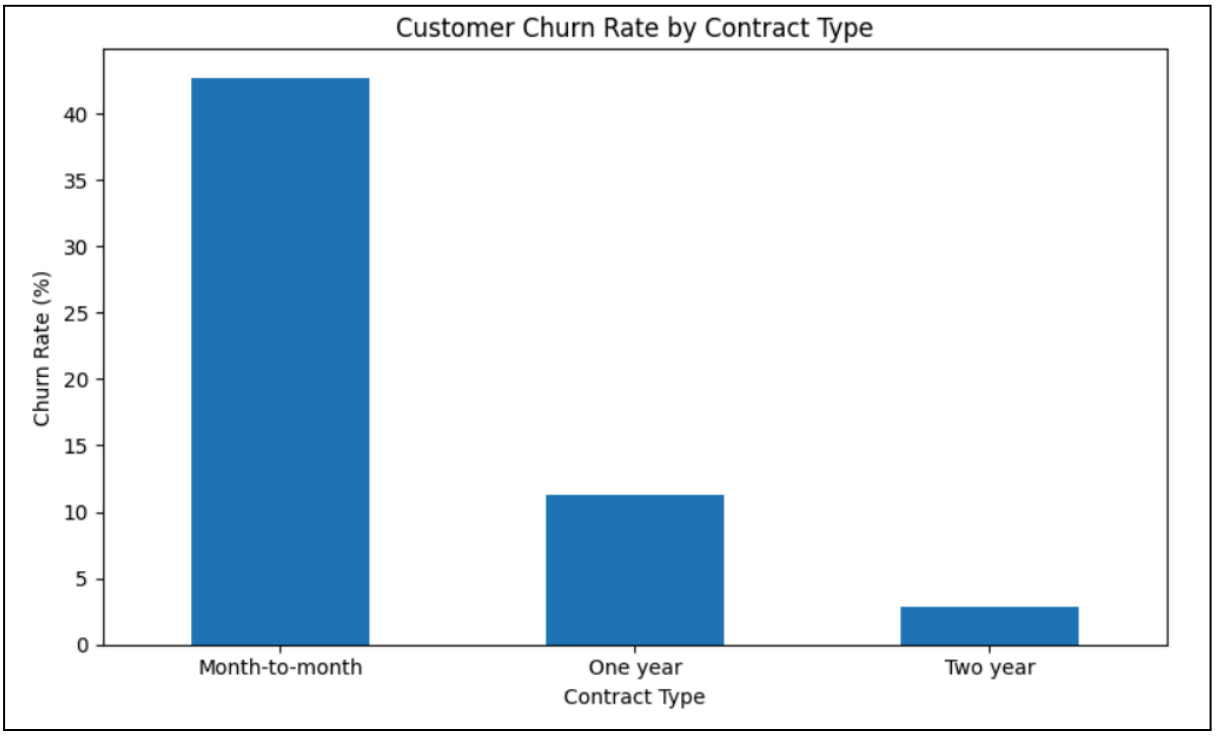
Customers who churned had average monthly charges of approximately 74.44, compared with approximately 61.27 for customers who stayed. The currency unit is not explicitly specified in the dataset.

Business Insight

Higher monthly charges may be associated with customer churn. Reviewing pricing plans and communicating service value may help improve customer satisfaction.

2.4 Customer Churn by Contract Type

Graph: Churn Rate by Contract Type



Interpretation

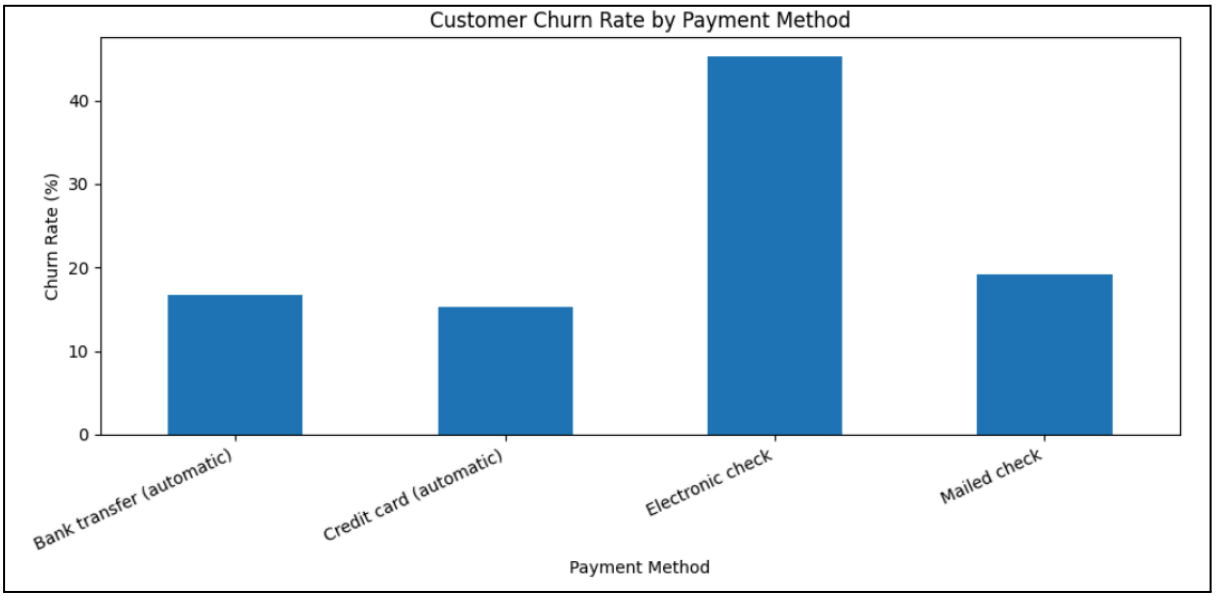
Month-to-month customers had the highest churn rate at approximately 42.71%. The churn rates for one-year and two-year contracts were approximately 11.27% and 2.83%, respectively.

Business Insight

Customers with shorter contract commitments are more likely to leave. Businesses could offer suitable long-term plans, loyalty benefits, and incentives to encourage continued subscriptions.

2.5 Customer Churn by Payment Method

Graph: Churn Rate by Payment Method



Interpretation

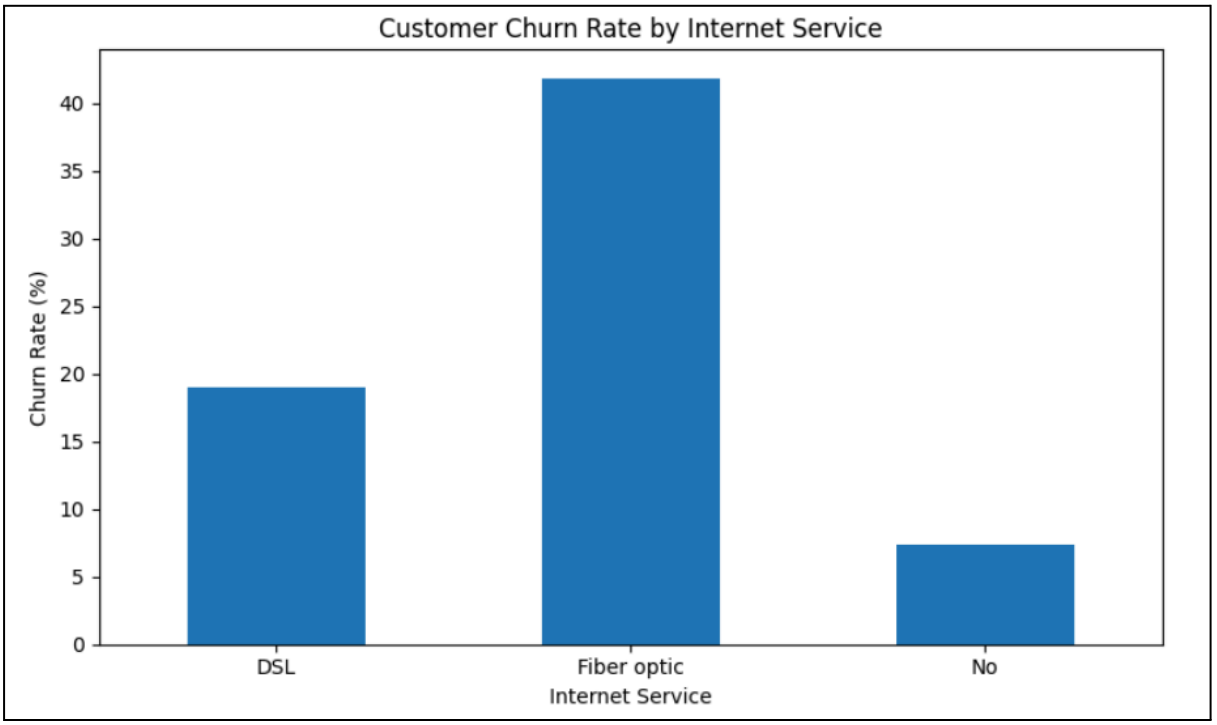
Electronic check users had the highest churn rate at approximately 45.29%. Customers using automatic bank transfer and automatic credit card payments had lower churn rates of approximately 16.71% and 15.24%, respectively.

Business Insight

The payment method is associated with different churn levels. Businesses could encourage convenient automatic payment options while investigating other reasons for the high churn among electronic check users.

2.6 Customer Churn by Internet Service

Graph: Churn Rate by Internet Service



Interpretation

Customers using fiber-optic internet had a churn rate of approximately 41.89%, compared with 18.96% for DSL customers. Customers without internet service had a churn rate of approximately 7.40%.

Business Insight

The higher churn rate among fiber-optic customers suggests that service quality, pricing, customer expectations, and competition should be investigated further.

3. Key Business Insights

The analysis of the Telco Customer Churn dataset revealed several important findings that can help the company understand customer behavior and improve retention.

3.1 Overall Customer Churn : The overall churn rate is approximately 26.54%, indicating that a considerable proportion of customers have discontinued the company's services. This highlights the importance of identifying at-risk customers and improving retention efforts.

3.2 Contract Type : Month-to-month customers have a substantially higher churn rate than customers on one-year or two-year contracts. Shorter contractual commitments may make it easier for customers to switch providers.

3.3 Customer Tenure : Customers who churned had a considerably shorter average tenure than retained customers. This suggests that the early stages of the customer relationship are important opportunities for improving retention.

3.4 Monthly Charges : Customers who churned had higher average monthly charges than those who stayed. Pricing, affordability, perceived value, and service expectations may be relevant factors to investigate.

3.5 Payment Method : Electronic check users showed the highest churn rate among the payment methods analyzed. Automatic payment users showed lower churn rates, suggesting an opportunity to investigate payment convenience and the customer experience.

3.6 Internet Service : Fiber-optic customers had a higher churn rate than DSL customers. The company should investigate whether pricing, service reliability, customer expectations, or competitive alternatives contribute to this pattern.

3.7 Overall Business Implication : The findings suggest that churn is not evenly distributed across customer groups. Contract type, tenure, monthly charges, payment method, and internet service are useful attributes for identifying groups that may require additional attention.

4. Business Recommendations

Based on the findings from the customer churn analysis, the following recommendations are proposed to improve customer retention and support data-driven business decisions.

4.1 Improve Early Customer Engagement : Develop an onboarding program for new customers that explains service features, billing procedures, and available support options. Follow up with customers during their initial months to identify and resolve issues before they decide to leave.

4.2 Encourage Long-Term Contracts

Offer appropriate incentives, loyalty rewards, or discounts to month-to-month customers who choose one-year or two-year contracts. These offers should be designed to provide value to customers while maintaining business profitability.

4.3 Review Pricing and Service Value

Evaluate monthly pricing plans, additional charges, and the value provided by each subscription. Consider offering flexible plans that better match different customer budgets and usage requirements.

4.4 Investigate Fiber-Optic Customer Churn

Examine customer feedback, service reliability, pricing, installation experiences, and support requests among fiber-optic users. Addressing any identified service or expectation gaps may help improve retention.

4.5 Promote Convenient Payment Options

Encourage customers to consider automatic payment methods by clearly explaining their convenience and available benefits. At the same time, investigate whether electronic check users experience billing difficulties or other issues that may contribute to churn.

4.6 Strengthen Customer Support

Improve access to technical assistance, billing support, and complaint resolution. Proactive communication and timely problem-solving may help strengthen customer trust and satisfaction.

4.7 Monitor Churn Through Dashboards

Use the Power BI dashboard to monitor churn rates, customer counts, contract types, payment methods, monthly charges, and tenure. Regular monitoring can help identify changes in customer behavior and evaluate the effectiveness of retention initiatives.

4.8 Measure the Results of Retention Strategies

Track churn rate, customer retention rate, customer satisfaction, and the performance of retention campaigns over time. Use these measures to determine which strategies provide the greatest improvement.

Expected Outcome :

These recommendations aim to reduce avoidable customer losses, improve customer satisfaction, strengthen long-term relationships, and support better business decisions. Their effectiveness should be evaluated using future customer data.

5. Power BI Dashboard (Tool Used : Microsoft Power BI Desktop)

Dashboard Overview

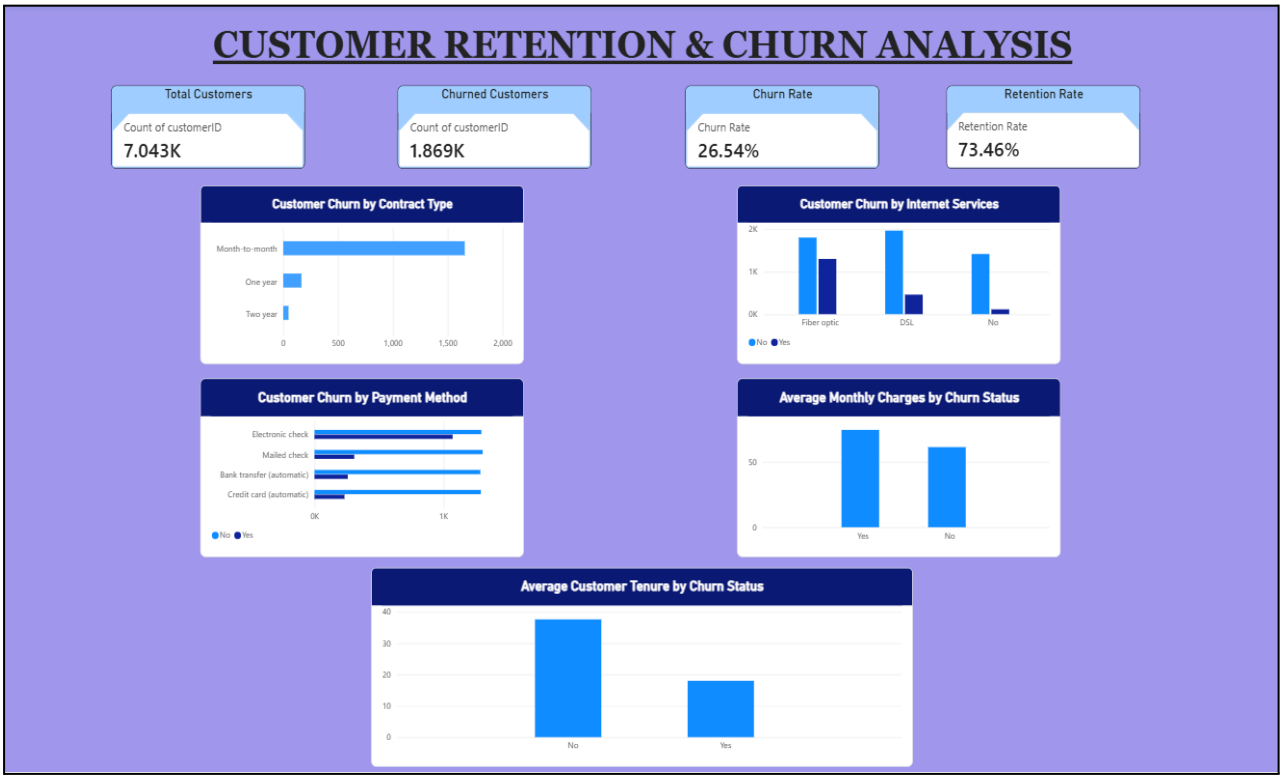
A Power BI dashboard report page was developed to present the results of the customer retention and churn analysis in a clear and visually accessible format. It combines key performance indicators (KPIs) and charts to help users understand customer behavior and identify important churn patterns.

The dashboard includes four KPI cards:

- Total Customers
- Churned Customers
- Churn Rate
- Retention Rate

It also includes five charts that analyze customer churn by contract type, internet service, and payment method, along with comparisons of average monthly charges and average customer tenure by churn status.

Figure : Customer Retention & Churn Analysis Dashboard



Key Dashboard Insights

The dashboard provides the following key insights:

- **Customer Overview:** The dataset contains 7,043 customers, including 1,869 customers who churned.
- **Churn and Retention:** The churn rate is approximately 26.54%, while the retention rate is approximately 73.46%.
- **Contract Type:** Month-to-month contracts show a higher churn rate than longer-term contracts.
- **Internet Service:** Fiber-optic customers show a higher churn rate than DSL customers.
- **Payment Method:** Electronic check users have the highest churn rate among the payment methods analyzed.
- **Monthly Charges:** Customers who churned had higher average monthly charges than retained customers.
- **Customer Tenure:** Customers who churned had a shorter average tenure than those who stayed.

Dashboard Benefits

The Power BI dashboard brings important customer metrics and churn patterns together in one report page. It makes the results easier to understand, supports comparison between customer groups, and helps decision-makers identify areas that may need further investigation or targeted retention efforts.

6. Conclusion

The Customer Retention and Churn Analysis project provided useful insights into customer behavior using the Telco Customer Churn dataset. Through data cleaning, exploratory data analysis, and visualization in Python, the project identified important patterns related to contract type, customer tenure, monthly charges, payment methods, and internet services.

The analysis showed that approximately 26.54% of customers had churned, while 73.46% remained with the company. Month-to-month customers, electronic check users, and fiber-optic customers had comparatively higher churn rates. In addition, customers who churned had higher average monthly charges and shorter average tenure than retained customers.

The Power BI dashboard presented these findings through KPI cards and visualizations, making the results easier to interpret and communicate. The insights can help businesses identify customer groups that may require additional attention and develop targeted retention strategies.

Overall, this project strengthened practical skills in data cleaning, exploratory data analysis, data visualization, and dashboard development. It also demonstrated how data analytics can support business understanding and more informed customer retention decisions.

The findings describe associations within the available dataset rather than proving the causes of customer churn. Further investigation, customer feedback, and ongoing monitoring would help validate the findings and evaluate the effectiveness of future retention strategies.